Category Theory for Mathematical Physics — Exercise Sheet 13

Boundaries, defects and non-invertible symmetries

Variant: Hints and answers

Conventions as in Lecture 13: the toric code has $A_s = \prod X$ on the edges at a vertex and $B_p = \prod Z$ around a face; $e = (0, \chi_-)$, $m = (1, 1)$, $\varepsilon = e \otimes m$ (here ε is the fermion), and its anyons are as in Theorem 12.8. An algebra is (L, μ, ι) and $d(L)$ its dimension. On the Ising chain of n sites, $\eta = \prod_j X_j$, and $\mathcal{T}$ is the translation, $\mathcal{T} O_j \mathcal{T}^{-1} = O_{j+1}$ (not the modular T); D is the Kramers–Wannier operator of § 13.3, and indices are mod n. $\zeta = e^{2\pi i/3}$, $\varphi = (1 + \sqrt{5})/2$, and ω is a 3-cocycle. In Exercise 6, $\varepsilon(b, c)$ is instead Sheet 3's carry, $\zeta_4 = i$, and ψ is a 2-cochain.

Exercise 1 (The toric-code boundaries on the lattice). (i) Draw a piece of the lattice with a smooth boundary on top and a rough one below. On it draw a string of Z's from a bulk vertex s down to the end of a dangling edge, and a string of X's on the edges crossed by a dual path from a bulk face p up through the smooth boundary. Show that each string violates only the term at its bulk end.

(ii) Show that a Z-string from s that ends at a vertex of the smooth boundary violates the star there too, and that an X-string from p that leaves through the rough boundary violates a three-edge plaquette. What does ε, brought to each boundary, leave behind?

(iii) The *dual lattice* has a vertex in each face and an edge across each edge. Show that passing to it, and exchanging X with Z, turns stars into plaquettes and back, the smooth boundary into a rough one, and e into m.

Hint. (i) A Z anticommutes with A_s exactly when its edge ends at s. (iii) What is the dual edge of an edge that lies in only one face?

Answer. (i) Each interior vertex (face) of the path meets two of its edges; the dangling end and the outer side have no term. (ii) The boundary star (three edges) meets one edge of the string, the truncated plaquette one; ε leaves e on the smooth boundary and m on the rough one. (iii) An edge in one face becomes a dangling dual edge, so smooth becomes rough and e (a star) becomes m (a plaquette).

Exercise 2 (Lagrangian algebras). (i) The double semion $\mathcal{Z}(\mathrm{Vec}^{\omega}_{\mathbb{Z}_2})$, $\omega(1,1,1) = -1$, has the four simple objects $(\delta_0, \pm 1)$, $(\delta_1, \pm i)$ (Sheet 12, Exercise 6; spins as in Lecture 12), all invertible of dimension 1, with spins $1, 1, i, -i$ (the self-braidings). Let $b = (\delta_0, -1)$. Show that $b \otimes b \cong 1$, and, as in the proof of Theorem 13.3, that the only Lagrangian algebra is $1 \oplus b$. How does this match Theorem 13.4 and the module categories over $\mathrm{Vec}^{\omega}_{\mathbb{Z}_2}$ (Lecture 11)?

(ii) In $\mathcal{Z}(\mathrm{Fib}) \simeq \mathrm{Fib} \boxtimes \mathrm{Fib}^{\mathrm{rev}}$ the simple objects 1, $\tau \boxtimes 1$, $1 \boxtimes \tau$, $\tau \boxtimes \tau$ have $d = 1, \varphi, \varphi, \varphi^2$ (Sheet 12, Exercise 5). Show that a connected L with $d(L)^2 = \dim \mathcal{Z}(\mathrm{Fib})$ is $1 \oplus (\tau \boxtimes \tau)$ as an object. With Sheet 11, Exercise 4(iii), conclude that $\mathcal{Z}(\mathrm{Fib})$ has exactly one Lagrangian algebra, and, by Theorem 13.5 and the physical principle of Lecture 13, exactly one gapped boundary.

Hint. (i) In degree 0 every associator is 1. If $X \otimes X \not\cong 1$, what is the product on $X \otimes X$? (ii) $\varphi^2 = \varphi + 1$, and $1, \varphi$ are linearly independent over $\mathbb{Q}$.

Answer. (i) $b \otimes b = (\delta_0, (-1)^2) = 1$; $L = 1 \oplus X$ needs $X \otimes X \cong 1$ and $\theta_X = 1$: only b. One Lagrangian algebra, one module category (the regular one). (ii) $L = 1 \oplus (\tau \boxtimes \tau)$, forced by d (φ is irrational); one module category, so one L and one boundary.

Exercise 3 (Condensing e). Let $L = 1 \oplus e$ in $\mathcal{D} = \mathcal{Z}(\mathrm{Vec}_{\mathbb{Z}_2})$, with the product of Theorem 13.3. A right L-module (M, ρ) is *local* if $\rho \circ c_{L,M} \circ c_{M,L} = \rho$. You may use: every L-module is a direct summand of a free one, $X \otimes L$ with $\rho = \mathrm{id}_X \otimes \mu$ (L is separable); the local modules form a braided fusion category $\mathcal{D}_L^{\mathrm{loc}}$, the phase left after condensing L, with unit L; and, as L is commutative, $X \otimes L$ with X simple is local exactly when the double braiding $c_{L,X} \circ c_{X,L}$ is the identity of $X \otimes L$.

(i) Show that $e \otimes L \cong L$ and $\varepsilon \otimes L \cong m \otimes L$ as L-modules, and that L and $m \otimes L$ are simple modules.

(ii) Show that L is local and $m \otimes L$ is not.

(iii) Conclude that $\mathcal{D}_L^{\mathrm{loc}} \simeq \mathrm{Vect}_{\mathrm{fd}}$: condensing e leaves the trivial phase. What happens to m and ε?

Hint. (i) Multiply by e inside L. A submodule of L is a subobject closed under the action. (ii) Use the double braidings of Sheet 12, Exercise 1.

Answer. (i) μ restricted to $e \otimes L$ is an isomorphism of modules; neither 1 nor e alone is closed under the action. (ii) e braids trivially with $1, e$; m has double braiding -1 with e. (iii) One simple local module, the unit L; m and ε are confined.

Exercise 4 (The four boundaries of $\mathcal{Z}(\mathrm{Vec}_{S_3})$). Write t for the class of transpositions and u for that of 3-cycles, and the eight anyons of Sheet 12, Exercise 3, as $1_{\mathbf{1}}, 1_{\mathrm{sgn}}, 1_{\mathrm{std}}$ (class $\{1\}$), $t_\pm$ and $u_1, u_\zeta, u_{\zeta^2}$ (the character of the centraliser at the class), with dimensions $1, 1, 2, 3, 3, 2, 2, 2$ and spins $1, 1, 1, 1, -1, 1, \zeta, \zeta^2$.

(i) Explain why $\mathcal{Z}(\mathrm{Vec}_{S_3})$ has exactly four Lagrangian algebras, and, by the physical principle, four gapped boundaries.

(ii) You may use (Beigi, Shor and Whalen, arXiv:1006.5479, §4.1, eq. (19)) that the boundary built from $\mathrm{Mod}(\mathbb{C}[H])$, $H \leq S_3$, condenses the anyons of the $D(S_3)$-module A_H with basis the pairs (xH, g), $g \in xHx^{-1}$, of flux g, on which $k \in S_3$ acts by $k \cdot (xH, g) = (kxH, kgk^{-1})$; the multiplicity of $([g], \pi)$ in A_H is that of π in the permutation representation of the centraliser of g on the pairs of flux g. Decompose A_H for the four classes of H.

(iii) Check that each A_H has dimension $6 = \sqrt{\dim \mathcal{Z}(\mathrm{Vec}_{S_3})}$ and only summands of spin 1. Which boundary condenses only pure charges, and which only pure fluxes?

Hint. (i) Sheet 11, Exercise 3. (ii) The pairs of flux g correspond to the cosets xH with $x^{-1}gx \in H$. For $H = \mathbb{Z}_3$, which is normal, the centraliser of a 3-cycle fixes both cosets.

Answer. (i) Four indecomposable module categories, by Theorem 13.4. (ii) $H = 1$: $1_{\mathbf{1}} + 1_{\mathrm{sgn}} + 2 \cdot 1_{\mathrm{std}}$; $\mathbb{Z}_2$: $1_{\mathbf{1}} + 1_{\mathrm{std}} + t_+$; $\mathbb{Z}_3$: $1_{\mathbf{1}} + 1_{\mathrm{sgn}} + 2u_1$; S_3: $1_{\mathbf{1}} + t_+ + u_1$. (iii) $1+1+4 = 1+2+3 = 1+1+4 = 1+3+2 = 6$; $H = 1$ charges, $H = S_3$ fluxes.

Exercise 5 (Kramers–Wannier on the ring). Use the relations $DX_j = Z_{j-1}Z_jD$, $DZ_{j-1}Z_j = X_{j-1}D$ and $\eta D = D$ of § 13.3, on a ring of $n \geq 3$ sites.

(i) Show that $D^2X_j = X_{j-1}D^2$ and $D^2Z_{j-1}Z_j = Z_{j-2}Z_{j-1}D^2$, and that $\mathcal{T}D^2$ commutes with every X_j and every $Z_{j-1}Z_j$.

(ii) Show that an operator commuting with every X_j and every $Z_{j-1}Z_j$ is $a + b\eta$ for numbers a, b.

(iii) Deduce that $D^2 = a(1+\eta)\mathcal{T}^{-1}$ (Seiberg, Seifnashri and Shao normalise D so that $a = 1$). Why is this not the fusion rule $N \otimes N = 1 \oplus \eta$ of a fusion category?

(iv) In *Integrability*, Lecture 18, the chain is $H = -J\sum_k \sigma^x_k\sigma^x_{k+1} - h\sum_k \sigma^z_k$, with parity $\Pi = \prod_k \sigma^z_k$ (its site written k here). Write the Kramers–Wannier relations and D^2 in those letters. Shao uses $D_{\text{Shao}} = D^\dagger/\sqrt{2}$, with D normalised as in (iii), and his T_{Ising} moves O_j to O_{j+1}: compute D^2_{Shao}.

Hint. (i) Apply the two relations in turn. (ii) Work in the basis of joint eigenvectors of the X_j; $Z_{j-1}Z_j$ flips two neighbouring eigenvalues. (iii) η commutes with $\mathcal{T}$. (iv) $\mathcal{T}$ is unitary.

Answer. (i) $D^2X_j = DZ_{j-1}Z_jD = X_{j-1}D^2$, and $\mathcal{T}X_{j-1} = X_j\mathcal{T}$. (ii) It is diagonal, $f(x)$, and constant on each parity class of x. (iii) $\mathcal{T}D^2 = a + b\eta$ and $\eta D^2 = D^2$ give $a = b$; $D^2 \neq 1 + \eta$, as $\mathcal{T}$ is not a scalar on the η-even states for $n \geq 3$. (iv) $D\sigma^z_k = \sigma^x_{k-1}\sigma^x_k D$, $D\sigma^x_{k-1}\sigma^x_k = \sigma^z_{k-1}D$, $D^2 = (1+\Pi)\mathcal{T}^{-1}$ (with the site written k); $D^2_{\text{Shao}} = \frac{1}{2}(1+\eta)\mathcal{T}$.

Exercise 6 (Full gauging and the anomaly). (i) For the cocycles ω_p of $\mathbb{Z}_4$ (Sheet 3), $p \in \{0,1,2,3\}$, use the invariant $Q_\omega(1) = \prod_{r=0}^{3}\omega(1,r,1)$, which is unchanged when ω is multiplied by a coboundary, to show that $\mathrm{Vec}^{\omega_p}_{\mathbb{Z}_4}$ is anomalous for $p \neq 0$. Which p allow a rank-one module category over the part on $\{0,2\}$ (Sheet 11, Exercise 2)?

(ii) Gauging all of Vec_{S_3}: using $H^2(S_3,\mathbb{C}^\times) = 0$ (given, as in Sheet 11, Exercise 3), how many choices of ψ are there, and what are the simple lines of the dual symmetry? Show that std has no inverse.

Hint. (i) $\omega_p(1,r,1) = \zeta_4^{\,p\,\varepsilon(r,1)}$. (ii) An invertible object has dimension 1.

Answer. (i) $Q_{\omega_p}(1) = i^p$, which is 1 only for $p = 0$; a rank-one module on $\{0,2\}$ exactly for $p = 0, 2$ ($\omega_p(2,2,2) = (-1)^p$). (ii) One ($H^2(S_3,\mathbb{C}^\times) = 0$); $\mathbf{1}$, sgn, std; $\dim \mathrm{std} = 2$.

Exercise 7 ⋆. *Ground states on a cylinder.* Put the toric code on a cylinder: circumference $W \geq 2$, vertex rows $y = 0,\dots,\ell$ ($\ell \geq 1$), and a smooth or rough boundary at each end (a rough end has dangling edges and three-edge plaquettes). You may use *Quantum Computing*'s counting: r independent commuting Pauli generators on n qubits, with $-\mathbb{1}$ not in the group they generate, fix a space of dimension 2^{n-r}.

(i) With a rough boundary below and a smooth one on top, count the qubits, stars and plaquettes, and show that the ground space is one-dimensional.

(ii) Do the same with both ends rough, and with both smooth: the ground space has dimension 2.

(iii) Compare with the number of anyons that condense on both boundaries, counted with multiplicity,

$$\dim \mathrm{Hom}(L_{\text{top}}, L_{\text{bottom}}),$$

for $L = 1 \oplus e$ (rough) and $1 \oplus m$ (smooth). Which string operators act on the two ground states when both ends are rough?

Hint. (i) As in *Quantum Computing*, a product of the stars at a set V of vertices is $\prod X$ over the edges with exactly one end in V. Which edges have only one end? (iii) An e-string from one boundary to the other, and an m-loop around the cylinder.

Answer. (i) $n = 2W(\ell+1)$ and $W(\ell+1)$ stars and plaquettes, all independent: $2^0 = 1$. (ii) Rough/rough: $n = W(2\ell+3)$, one relation (all plaquettes); smooth/smooth: $n = W(2\ell+1)$, one relation (all stars): 2^1. (iii) $1, 2, 2$; a Z-string between the boundaries and an X-loop around the cylinder, anticommuting.